CMAP Emme → GMNS/TAPLite Conversion — Quick Reference

Conformity cycle `c25q4`, scenario 100 (2019 base year) · PM period (Emme period 7, 4–6 pm) · 55,005 Emme links → 55,005 GMNS links · 3,649 zones · 27 Oct 2026

1 Inputs → code → outputs

INPUT	CODE	OUTPUT

`highwayshp_pmpk/emme_links.shp`
`highwayshp_pmpk/emme_nodes.shp`
`.dbf .shx .prj .cpg`
	`cmap_emme/emme_read.py`
	`cmap_emme/network.py`
		`network/PM/node.csv`
		`network/PM/link.csv`
		`network/link_geometry.csv`

`emmenet_highway/attribs_p7.txt` (`@speed @ftime @emcap @toll...`)	`cmap_emme/vdf.py` `cmap_emme/access.py`	capacity, free-flow, VDF, `allowed_use` `parameters/vdf_parameters.csv`
`c25q4_100/emmemat_c25q4_100.zip` (`.emx`, read in place)	`cmap_emme/demand.py`	`demand/PM/sov_PM.csv` ... (8 files) `parameters/mode_type_by_period.csv`
`emmenet_highway/network_p7.txt`	`cmap_emme/checks.py`	`checks/gate_report.csv` (cross-check only)

Driver `python code/main_convert.py --region region.cmap.yaml --periods PM` (`--skip-demand`: network only)
Code root `shp2gmns_replay_kit/code/src/cmap_emme/`; every CMAP number lives in `region.cmap.yaml`
Output root `shp2gmns_replay_kit/data/output/cmap/`; gates: 29 PASS, 0 FAIL, 17 INFO

2 Source tags used below

[G]	CMAP Travel Demand Model Data User Guide (c25q4), in the data package: CMAP Data Hub	[FT]	`Database/macros/Ftime_capacity.py` (`@ftime`, `@emcap`)
[A]	ON TO 2050 travel demand model appendix: cmap.illinois.gov (printed pages)	[TOD]	`Database/macros/TOD_network_prep.py` (VDF inputs ul1–ul3)
[VDF]	`Database/data/vdf_tod.in` (functions fd1–fd8 as run)	[TT]	`Database/macros/ttables.mac` (period trip tables)
[TF]	`Database/data/tod_factors.p7` (PM factors r19–r26)	[FL]	`Database/data/tod_flags.in` (airport and CBD zone flags)
[SOLA]	`Database/macros/SOLA_assignment.py` (toll as link cost)	[Y]	`region.cmap.yaml`
[m]	measured from the c25q4 PM files		

Code files are from `CMAP-REPOS/cmap_trip-based_model`, commit `e52b318` (2026-04-30); copies in `data/input/cmap/docs/cmap_code/`.

3 The input files at a glance

Facts [m]	What it actually is
55,005 directed links, 20,190 nodes; no duplicate (i, j)	Emme PM-peak network exported to shapefile [G]
Zones 1–3649, contiguous, `ISZONE = 1`; node < 5000 is a centroid	Zones already come first: **no renumbering**
CRS NAD27 / Illinois East, US survey feet (EPSG:26771) → EPSG:4326	[A] T19 p.48; model area includes Winnebago Co., NW Indiana, SE Wisconsin
Shapefile has **no** capacity, speed or free-flow time	These are in `attribs_p7.txt`; joined on (i, j): 55,005 / 55,005
`DATA1` = `@vadt`; `TIMAU` = `timau` on every link	CMAP's assigned PM volume and congested time

Used	shapefile `ID INODE JNODE LENGTH MODES LANES VDF DATA1 TIMAU`; attribs `@speed @ftime @emcap @toll @tipid @width @parkl @sigic @vadt timau`
Ignored	`TYPE` (always 1), `VOLAU` / `VOLAD` (**not volumes**: median 3,648), `DATA2 DATA3` (all 0), `@avelw` (same kind as `@width`), `emme_turns.dbf` (turns, not converted)

4 Code tables

`VDF` — **volume-delay function code** [A] T18 p.47, [G]

Code	Meaning	Links	PM VMT
1	Arterial street	41,861	16,836,841
2	Freeway	1,998	10,565,352
3	Freeway → arterial ramp	1,992	364,490
4	Expressway	930	1,436,129
5	Freeway → freeway ramp	664	338,540
6	Zone centroid connector	7,264	3,893,909
7	Toll collection link	188	67,954
8	Metered entrance ramp	108	19,478
	total	**55,005**	**33,522,694**

`MODES` **letters** [A] T18 p.47

A primary auto, S SOV, H HOV, T general truck, b B-plate truck, l light, m medium, h heavy.

B-plate [A] p.111: vans, pickups and SUVs under 8,000 lb carrying "B" plates; only their commercial use (personal use is in household trips).

Mode codes (= OD vehicle classes) [Y] access: A → `sov;apv`, S → `sov`, H → `hov2;hov3`, b → `btrk`, l → `ltrk`, m → `mtrk`, h → `htrk`; T adds nothing.

`MODES` → `allowed_use` (NVTA convention: mode codes, `;`-separated)

MODES	`allowed_use` written	Links	Meaning
`ASHThmlb`	`sov;hov2;hov3;apv;btrk;ltrk;mtrk;htrk`	50,274	all vehicles
`ASHTb`	`sov;hov2;hov3;apv;btrk`	1,995	autos and B-plate only
`ASH`	`sov;hov2;hov3;apv`	1,518	no trucks
`ASHTlb`	`sov;hov2;hov3;apv;btrk;ltrk`	875	no medium or heavy
`ASHTmlb`	`sov;hov2;hov3;apv;btrk;ltrk;mtrk`	343	no heavy

Every PM link carries A, S and H (no HOV-only lanes); no link is `closed`.

5 Master lookup: free-flow time, capacity, VDF

Free-flow time `@ftime` [FT]: `@speed` is rounded **up** to a bin, then `@ftime` = `length` / bin speed × 60. The converter uses `@ftime` as given; recomputed from `@speed` it matches on 100% of VDF 1–6, 8 links [m].

VDF	Speed bins (mph)	Extra	`@ftime`≠`LENGTH/@speed` [m]
2, 4	40, 45, 50, 55, 60, 65 (≤40→40, >60→65)	—	520 (e.g. fwy 70→65; xwy 25–35→40)
1	15, 20, . . . , 50, 55 (>50→55)	signal delay later, in fd1	0
3, 5, 8	15, 20, . . . , 45, 50 (>45→50)	VDF 8: +0.083 min (5-s meter delay)	220 (ramps 55–65→50; VDF 8)
6	—	`@ftime` = `length` × 3 (20 mph)	—
7	—	posted speed of the incoming link	1

Capacity `@emcap` (veh/h/**lane**, LOS E) [FT], [G]

VDF	Facility	Median	Range	Values
1	arterial	790	288–1,449	254
2	freeway	2,000	1,140–2,000	7
3	fwy→art ramp	1,450	620–1,600	8
4	expressway	1,700	1,020–2,000	16
5	fwy→fwy ramp	1,450	1,060–1,600	6
8	metered ramp	720	720	1
6, 7	connector, toll	0 in CMAP → 99,999 placeholder		

Freeway `@emcap` is 2,000 for every lane count 1–7, so it is per lane. Arterial capacity depends on the downstream intersection (approaches, lane width, functional class) [FT] l.250–330.

VDF inputs as CMAP runs them [TOD], [VDF]

`ul1`	`@ftime`
`ul2`	0.75 × `@emcap` × `lanes` × H on **VDF 1–5**; `@emcap` × `lanes` × H on VDF 8; PM $H = 2$
`ul3`	signal cycle and g/C (VDF 1, 3); **not exported**
v	`volau + el2`: vehicle equivalents plus bus veq

[A] p.123 names the 0.75 only for arterials; the code applies it to VDF 1–5, and the code is followed.

Functions [VDF] and reproduction of CMAP's `timau` with v = `DATA1` [m]

VDF	Name	fd (as run)	Median err.	TAPLite form
1, 3	arterial, ramp	$u_1[1+0.15(v/u_2)^4]$ + Webster uniform + incremental delay from u_3, capped at one cycle	−21.8%, −0.04%	open
2	freeway	$[1+0.15\min(v/u_2,1)]\,(u_1/1.15)\,[1+0.15(v/(u_2+300))^8]$	−9.5%	open
4	expressway	$[1+0.15\min(v/u_2,1)]\,u_1\,[1+0.15(v/u_2)^8]$	−1.3%	open
5	fwy→fwy ramp	$u_1[1+0.15(v/u_2)^8]$	−0.04%	open
6	connector	`length` × 3	0.0%	type 0, $\alpha = 0$
7	toll link	u_1	0.0%	type 0, $\alpha = 0$
8	metered ramp	$u_1[1+0.15(v/u_2)^{10}]$	−0.04%	open

The remaining error is the volume: `DATA1` is vehicles, CMAP's v is vehicle and bus equivalents. VDF 1 misses the signal delay (`@cycle`, `@gc` not exported). The TAPLite form for VDF 1–5, 8 is chosen in the replay step, so `vdf_type`, `vdf_alpha`, `vdf_beta` are blank on those links.

6 `node.csv`

Column	From	Rule [Y], network.py
`node_id`	`emme_nodes.dbf ID`	Emme node number, unchanged
`zone_id`	`ISZONE`	= `node_id` where `ISZONE = 1` (1–3649); else blank
`x_coord`	`X` (US survey ft)	EPSG:26771 → EPSG:4326, degrees
`y_coord`	`Y` (US survey ft)	same
`node_type`	`ISZONE`	`centroid` or `road`
`cmap_label`	`LABEL`	as given

20,190 rows; 3,649 centroids. Reprojection: pyproj when installed; otherwise a built-in inverse Transverse Mercator (Clarke 1866) plus Molodensky NAD27 → WGS84, checked against a pyproj-based CMAP–OSM match: median 2.11 m vs 2.12 m [m].

7 `link.csv`

Column	Formula	Unit	Note
`link_id`	row number 1..N	—	shapefile order
`from_node_id`	`INODE`	—	
`to_node_id`	`JNODE`	—	
`dir_flag`	1	—	Emme links are directed
`length`	`LENGTH` × 1609.344	**meter**	source is miles [A] T18
`lanes`	`LANES`	—	≥ 1 on every link
`capacity`	`@emcap`; 99,999 on VDF 6, 7	**veh/h /lane**	§5; placeholder §11
`free_speed`	`vdf_free_speed_mph` × 1.609344	**kph**	from `@ftime`, not `@speed`
`toll`	`@toll`	**dollar /link**	CMAP adds it unchanged as link cost [SOLA]
`link_type`	`VDF`	—	= `vdf_code`
`allowed_use`	`MODES` → mode codes	—	§4
`vdf_type`	0 on VDF 6, 7; blank otherwise	—	replay step
`vdf_alpha`	0 on VDF 6, 7; blank otherwise	—	
`vdf_beta`	1 on VDF 6, 7; blank otherwise	—	
`vdf_plf`	1.0	—	period cap = `@emcap`×lanes×2 [G]
`ref_volume`	`DATA1` (= `@vadt`)	**veh / 2-h period**	CMAP assigned PM volume, not per hour
`ref_time`	`TIMAU` (= `timau`)	**minute**	VDF time at that volume, §7 note
`vdf_free_speed_mph`	`LENGTH` / `@ftime` × 60	**mph**	binned speed [FT]
`vdf_length_mi`	`LENGTH`	**mile**	
`vdf_fftt`	`@ftime`	**minute**	as given [FT]
`org_link_id`	`ID` (`"i-j"`)	—	
`vdf_code`	`VDF`	—	1–8
`facility_type`	name of the VDF code	—	§4
`capacity_source`	`cmap_emcap` / `placeholder_no_constraint`	—	filter before D/C
`hourly_link_capacity`	`capacity` × `lanes`	veh/h/link	
`tipid width parkl sigic`	`@tipid @width @parkl @sigic`	—, ft, count, 0/1	CMAP names without `@`
`timau`	`TIMAU`	minute	same as `ref_time`, CMAP's name

Unit contract (kit convention, `dtalite4cube/unit_contract.py`)

`length`	= `vdf_length_mi` × 1609.344
`free_speed`	= `vdf_free_speed_mph` × 1.609344
`vdf_fftt`	= 60×`vdf_length_mi`/`vdf_free_speed_mph`

What `ref_time` / `timau` is [G], [VDF]

CMAP runs a *static* equilibrium assignment per period. Every link ends with one volume (`DATA1`, whole 2 h) and the VDF time at that volume. One time for 4–6 pm at the period-average flow rate: no change within the period, no queue spillback. A model output ("raw model results" [G]). 2,823 links carry zero volume and keep free-flow time.

8 Demand

Built as CMAP's `ttables.mac` builds the period-7 tables [TT]; factors [TF] = [G] "Daily truck and external trip time-of-day conversion", TOD7. Each `.emx` is float32, 3649 × 3649, no header.

mode	File	Rule [TT] line	PM veh	Rows	Kept	MB
sov	sov_PM.csv	mf417 + mf427 + mf437 + 0.155×mf8 (l.392–444)	2,393,513	818,885	100.0%	10.1
hov2	hov2_PM.csv	mf447	513,112	342,932	100.0%	4.3
hov3	hov3_PM.csv	mf457	141,612	204,014	100.0%	3.6
apv	apv_PM.csv	0.155×mf10, airport rule (l.464–465)	11,839	26,944	99.9%	0.5
btrk	btrk_PM.csv	0.176×mf4 (l.601)	307,563	1,338,781	96.5%	25.1
ltrk	ltrk_PM.csv	0.176×mf5 (l.626)	51,627	718,863	71.9%	13.7
mtrk	mtrk_PM.csv	0.176×mf6 (l.658)	48,066	664,820	70.9%	12.7
htrk	htrk_PM.csv	0.103×mf7 + 0.103×mf9 (l.684–685)	60,398	659,707	87.6%	12.7
		total	**3,527,730**	**4,774,946**	**98.7%**	**82.7**

Format	`o_zone_id,d_zone_id,volume`; zone ids 1–3649 = centroid `node_id`; **vehicles per 2-h period**; intrazonal cells kept.
Floor	cells ≤ 0.01 not written, **not rescaled** (first version); kept share per mode in `parameters/demand_summary.csv`.
PM autos	mf417–mf457 are PM vehicle trips from CMAP's demand model (`mf41y..mf45y`, $y = 7$); PM2 = period 7 (`setup_transit_asmt_3_TOD_transit_demand.py`). mf417+mf427+mf437 = 2,351,437 = `trip_roster` PM2 mode 1 [m].
Airport	[FL]: O'Hare 123–127, 129, 130; Midway 494; 1733, 1734, 1969, 2907 flagged ≥ 2; CBD 1–47 flagged 1. Kept: to an airport, or from an airport to an unflagged zone. CMAP folds these into SOV; here `apv` (our choice).
Trucks	CMAP converts to veq inside `ttables.mac` (medium ×2, heavy ×3); here vehicles, with the factor in `pce`. mf9 into heavy: "externals, which are assumed to be primarily heavy trucks" [TT] l.676.
Not included	park-and-ride first legs (mf59, built inside CMAP's run); bus veq (a link preload, replay step).
Do not use	emmemat mf94–mf97, mf14–mf18 hold period 8 (the last assigned), e.g. mf14 = 0.152×mf4.

9 `mode_type.csv` values (`parameters/mode_type_by_period.csv`)

id	mode_type	vot	pce	occ	demand_file
1	sov	20.49	1	1	sov_PM.csv
2	hov2	20.25	1	2	hov2_PM.csv
3	hov3	31.50	1	3.36	hov3_PM.csv
4	apv	30.00	1	1	apv_PM.csv
5	btrk	24.00	1	1	btrk_PM.csv
6	ltrk	32.00	1	1	ltrk_PM.csv
7	mtrk	47.00	2	1	mtrk_PM.csv
8	htrk	52.00	3	1	htrk_PM.csv

`vot`	$/h, [TT] r100–r114 × perception factors. `sov`: PM-trip-weighted mean of VOT1 2.3 × 2.25, VOT2 7.2 × 2.25, VOT3 24 × 2.00 (externals split 0.10/0.45/0.45, r61–r63).
`pce`	medium 2, heavy 3: CMAP's veq [TT], [A] p.122.
`occ`	HOV3+ 3.36 [G]; others definitional.

One `sov` class replaces CMAP's three VOT classes; this matters only on the 198 tolled links.

10 Worked example: link `3634` → `11223`

Freeway leaving external zone 3634 (north edge of the model area, Wisconsin). All values are the actual file values [m].

1. Source rows

`ID`	3634-11223
`LENGTH`	0.58
`LANES`	3
`VDF`	2
`MODES`	`ASHThmlb`
`DATA1`	4605.3213
`TIMAU`	0.521654
`@speed`	65
`@ftime`	0.535385
`@emcap`	2000
`@toll`	0

2. `node.csv`

`node_id`	`zone_id`	`x_coord`	`y_coord`	`node_type`
3634	3634	−87.920880	42.966647	centroid
11223	*blank*	−87.932119	42.965410	road

4. Demand — zone 3634 → 3636 (external → external, through trip)

`sov_PM.csv`	`3634,3636,402.734`	= 0.155× mf8 (2,598.285)
`htrk_PM.csv`	`3634,3636,114.696`	= 0.103× mf9 (1,113.551)
other modes	no row	mf417–457, mf4–7 are internal only

3. `link.csv`

Column	How	Column	How
`link_id`	3634 (row number)	`vdf_length_mi`	0.58
`from_node_id`	3634	`vdf_fftt`	0.535385 = 0.58/65 × 60
`to_node_id`	11223	`vdf_free_speed_mph`	65.0
`length`	0.58 × 1609.344 = 933.42	`free_speed`	65 × 1.609344 = 104.6073
`lanes`	3	`capacity`	2000 (`cmap_emcap`)
`link_type`	2 (freeway)	`hourly_link_capacity`	6000
`allowed_use`	`sov;hov2;hov3;apv;btrk;ltrk;mtrk;htrk`	`ref_volume`	4605.3213
`vdf_type/alpha/beta`	blank (replay step)	`ref_time`, `timau`	0.521654

fd2 check: $u_2 = 0.75{\times}2000{\times}3{\times}2 = 9{,}000$; $v/u_2 = 0.512$; $t = (1{+}0.15{\times}0.512){\times}(0.535385/1.15){\times}[1{+}0.15{\times}(4605.32/9300)^8] = 0.502$ min vs `TIMAU` 0.522 (−3.9%: CMAP's v counts trucks and buses as veq).

11 Gotchas

`@speed` ≠ speed in `@ftime`	`@ftime` uses `@speed` rounded up to a bin [FT]; 741 links differ (`checks/ftime_vs_posted_speed.csv`). `@speed` is not written.
0.75 on VDF 1–5	[TOD] applies it to all of VDF 1–5; [A] p.123 names arterials only. Code followed.
`/1.15` on freeways	applied inside fd2, not in `@ftime`.
`VOLAU` in the shapefile	not a volume (median 3,648). Use `DATA1`.
`@toll` is per link	[A] T1's per-mile rate is the master-network field, converted before export; CMAP adds `@toll` unchanged [SOLA]. 198 tolled links: 188 VDF 7 (median $0.70), 10 VDF 2 (median $0.535).
Connector capacity is 0	VDF 6 and 7 have `@emcap = 0`; 99,999 placeholder with $\alpha = 0$ (`region.cmap.yaml > vdf > no_capacity_links`, CMAP only; NVTA uses 3,150, SCAG its own file). 7,452 links.
Zone links ≠ connectors	32 road links (VDF 1, 2, 4) leave external zones 3633–3649; they keep `@emcap`. Connectors are found by VDF 6, not by touching a zone.
Toll links stay	in series on the tollways; deleting them drops 940 nodes and 1 zone from the connected network.
emmemat period tables	mf94–97, mf14–18 are period 8, not PM.
Trucks in veq	CMAP's tables are veq; here vehicles + `pce`.
Floor 0.01	drops 28–29% of light and medium trucks, 12% of heavy; not rescaled (first version).
Arterial delay	needs `@cycle`, `@gc`; not exported, so VDF 1 is −22%.
Base year	scenario 100 = 2019; `c25q4_200` (2025) has the same file structure.
No pyogrio / pyproj	built-in shapefile reader and reprojection are used; checked to 0.0 m median against pyproj.

12 Assignment test

The converted network and demand assign in TAPLite without changes (6 Nov 2026): PM, 5 classes (four car modes merged), BPR $\alpha = 0.15$, $\beta = 4$ on `@emcap` × lanes, no tolls, 10 iterations, gap 0.18%. Against CMAP's `ref_volume` and `timau`, all links: VMT 0.925, VHT 0.710, delay 0.109; link volume R^2 0.93 on road links. Results and the reading: `report/CMAP_progress_report.pdf`, sections 4–5. Inputs, run scripts and evaluation: `city-tensor/cmap_ta_bpr_allmodes`.